

16.1 Stages in the system life cycle

The life cycle of a computer system usually consists of five main stages, as shown in Figure 16.2. Each stage includes a range of tasks. Following the final stage, the system life cycle may repeat, with changes or additions being made to the system, such as taking in feedback from the end users.

- 1 **Analysis.** During the analysis stage, the current situation is researched and the requirements of the new system determined.
- 2 **Design.** Information from the analysis stage is used to examine the processing required and how data flows through the system. Inputs, outputs and data storage are also examined during the design stage.
- 3 **Development and testing.** Once the system has been designed, the required hardware may be purchased and/or the required software developed. Testing of the system takes place, with changes made to the system where required.
- 4 **Implementation.** In the implementation stage, the system begins to be used. This will include the development of user documentation. The system is also evaluated to ensure it is suitable for the intended use.

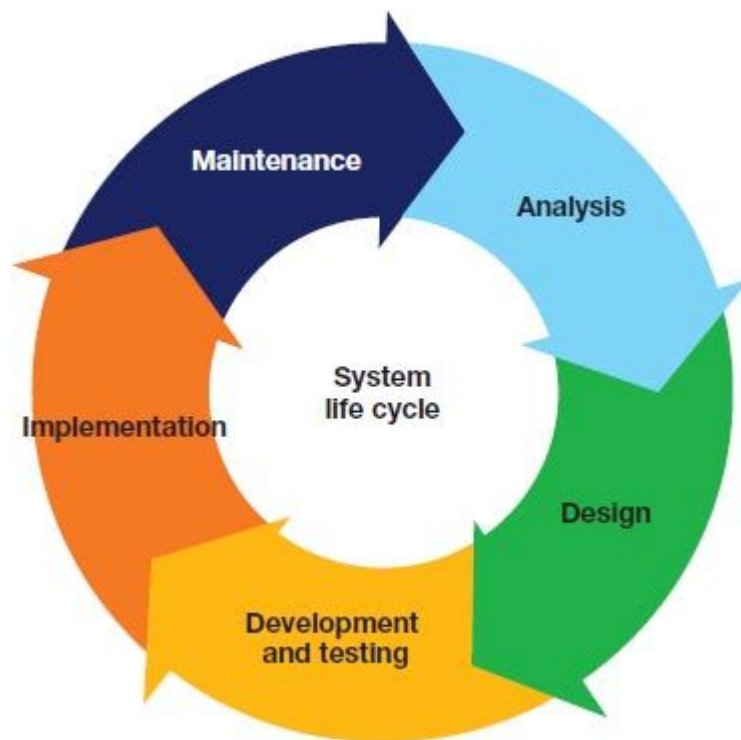


Figure 16.1: System life cycle.

- 5 **Maintenance.** Maintenance takes place to ensure the system continues to work as expected.

Stage 1: Analysis

Before a computer system can be set up or upgraded, the business or organisation first needs to carry out some research to understand the given situation, including how current

systems and processes work, and determine what the requirements are for the new system. This will involve **analysis**.

Methods of researching a given situation

The organisation can use a variety of methods to research a given situation.

Questionnaires

Questionnaires are used when information is required from a large number of users but it would be impractical to interview them all. A large number of users also means there is a large sample size for the results of the questionnaire to be analysed. Questionnaires may not be suitable when there are only a small number of users involved because there is not a large enough sample size to gauge opinion, and it would be quicker to conduct interviews. However, questionnaires could be used as an alternative to interviews if it is impossible to arrange an appointment time with a user or users. A disadvantage of questionnaires is that they do not allow the opportunity to ask the user to explain their choice of answers without contacting the user again.

Questions need to be asked in a way in which the required information can be obtained from users but also so that the responses can be easily analysed. This often means providing multiple-choice responses so that each response can be counted (see Figure 16.2). It is also important to ensure that the questionnaire does not take too long for users to complete, as users may not wish to complete the questionnaire, or answer all the questions, if it is too long.

WORKED EXAMPLE 16.01

During the analysis for a new school reports system, it is decided to find out from students, teachers and parents what information should be included in the report. One question that could be asked would be:

Please rate from 1 to 5 the importance of having the following information on the school report (1 is not important, 5 is very important):

- Attendance total half days
- Percentage score for each end of year exam
- Attendance percentage
- Average score for all exams
- Number of negative behavioural events
- Comment from subject teacher
- Number of positive praise events
- Targets from subject teacher
- Academic grade for each subject
- Comment from personal tutor
- Position/rank in class for each subject
- Target grade.

This question will allow the school to consider the importance given to each piece of information by students, teachers and parents. This will help to decide what

information should be included in the report.

Another way to ask this question would be:

Please list any information you would like to be included in the school report.

This question would make it difficult to quantify the responses and analyse the findings as each respondent may give very different answers. By providing a list, respondents are given a starting point.

A combination of multiple-choice questions, opinion ratings and open questions can be used within a questionnaire. A range of question types can provide a balance of quantitative analysis of closed questions (expecting yes/no answers) and a qualitative analysis of open questions, where users are able to suggest alternative ideas to those presented by the questionnaire. Questions should also be written in a way which does not threaten users and the way they currently do their work. Users should be given the opportunity to return their questionnaires anonymously, as more honest answers are likely to be given if it is not clear who has given the response.



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Figure 16.2: Completing a paper questionnaire.

Questionnaires can be completed online, which enables the results to be immediately stored and readily available for detailed analysis in the form of graphs and tables. Filters can be applied to the results and responses can be compared based on the answers given to another question. For example, a filter could be applied to compare the responses of all people who work part-time, compared with people who work full-time.

Interviews

Interviews involve a direct conversation between the person undertaking the analysis and the organisation (the client). Where there is a single end user or small group of end users, interviews are often used because questions can be asked of the users allowing a conversation to take place. This gives opportunities for follow-up questions. Even in large organisations, interviews may be used with key stakeholders or specific groups.

Questions for interviews should be selected to obtain the required information from the client or end users. The questions might vary depending on who is being interviewed. For example, if management are being interviewed, then the questions could focus on the

requirements of the organisation as a whole and the information that is required for decision making. If end users are being interviewed, then the questions could be aimed at finding out what the users need to make their jobs more efficient (see Figure 16.3). Interviews can also take place with groups of users or focus groups that represent user groups or customers.



Figure 16.3: End users discuss ways to make their jobs more efficient.

It can sometimes be difficult to find a time when client or end users are available, or to get everyone together in a focus group. Interviews can be a time-consuming process, particularly when there are many interviews to complete.

Honesty is important during interviews so that the interviewer can get an accurate picture of how tasks are completed. This can sometimes be difficult to achieve because end users may not want to admit to taking shortcuts in their tasks, or to not carrying out tasks to the best of their ability. In these situations, anonymous questionnaires can get more honest responses.

Observation

Observation involves the person that is doing the analysis watching the processes that take place within an organisation to find out how tasks are completed. This can involve sitting with users to understand the tasks they have to complete, with an opportunity to ask questions of the users to find out further information, which could form part of a **user requirements specification**. This can give a good understanding of the current input data, processing methods and output information. Other options can include visiting different parts of an office or other business locations throughout the day to see how tasks are undertaken.

One disadvantage of observations is that when users are being observed, they may do things differently from normal, or they may try to be more efficient, which does not give the observer a true picture of what is happening. The observer needs to be able to identify how long tasks genuinely take and any inefficiencies that could be reduced. By observing users directly, the person that is doing the analysis can get first-hand experience of any inefficiencies and can plan to overcome these. During observations, some users may not like being watched and this may lead to additional stress. The user may therefore not perform as they would do in normal work conditions. Although observations can take up a lot of time, it is often a good way to find out how an organisation works.

Document analysis

Documents used by an organisation can be analysed to understand how information within an organisation is currently being used.

UNPLUGGED ACTIVITY 16.01



Figure 16.4: Shop till receipt.

Examine the receipt in Figure 16.4.

- 1 Identify the output information that is included on the receipt.
- 2 Identify any calculations that are likely to have taken place on the receipt.
- 3 Identify which information is likely to be the same for all receipts and which information is likely to be different for each receipt.

It is important to look at examples of any documents that show data from an existing process, such as information output following a task, or to understand what data is being collected for input to a system. Sometimes it may be possible to identify processes that take place by looking at documents. It is also possible to estimate the amount of data that is likely to be required if the volume of documents is known.

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Document analysis should not be used on its own, but must be used with other analysis methods, because it is difficult to identify the processes just by looking at documents. Examination of the documents also only shows data that is currently used. It does not give an opportunity to find out what additional data an organisation might need, or what data the organisation does not need. This information could be found out by following up document analysis with interviews.

Advantages and disadvantages of research methods

Table 16.1 sets out the advantages and disadvantages of different research methods.

Method	Advantages	Disadvantages
Questionnaires	Efficient data collection from a large number of users. Users can submit responses anonymously.	Questionnaires can be time consuming to develop. Information provided can be limited.
Interviews	Enable a detailed analysis of users' views.	Time consuming to undertake.

Method	Advantages	Disadvantages
	Further questions can be asked to clarify responses.	Limited number of responses collected.
Observation	Enables tasks undertaken to be seen in person. Links between different processes can be identified.	Takes a long time to perform research. Users can behave differently when being watched.
Document analysis	Existing data and forms can be studied in detail. Requires limited involvement from users.	May not be clear how documents are used within the organisation. Other data required may not be identified.

Table 16.1: Advantages and disadvantages of the different methods of researching a given situation.

Content and purpose of specifications

Following research and analysis of the given situation, three types of specification are usually produced to understand and document the requirements of the system to be developed. They include user requirements specification, design specification and system specification. These documents help ensure the system meets the needs of the end users and other stakeholders.

User requirements specification

A user requirements specification sets out (specifies) what the organisation or client needs the system to do. The specification is written in consultation with the client, who will approve it (see Figure 16.5).

A user requirements specification should include:

- the purpose of the system
- the main objectives of the system
- data that must be output from the system (for example, invoices, sales reports)
- data that needs to be input to the system to generate the outputs, including any screens or data collection forms
- validation and verification that is needed for input data
- processes that need to take place to convert inputs into outputs or to store data
- data that needs to be stored
- functional requirements, such as performance measures
- deadlines for each milestone within the project (see [Chapter 15](#) for more information on project milestones).



Figure 16.5: The user requirements specification is written in close consultation with the client.

WORKED EXAMPLE 16.02

Here is an extract from a user requirements specification for a new local government website. The extract shows specific data that is required on the home page, in addition to what has been specified to be included on each page:

- Quick links
- Show list of links editable by content manager
- Initially to be:
 - How can I get involved? (go to Get involved)
 - How can I stand for council? (go to Elections)
 - When is the next steering group meeting? (go to Meetings)
 - When is the next information evening? (go to Key dates)
 - How do I report a problem? (go to Reporting)
 - When are the next elections? (go to Elections arrangements)
- News list:
 - Picture, news title, date (taken from list of news)
 - Button: view all news
 - To show latest four news articles
- What's on list:
 - Picture, event title with hyperlink, date and time (taken from List of events, Key dates and Meetings)
 - Button: view all events
 - Button: view all key dates
 - To show next four events.

System specification

A **system specification** lists all the software and hardware that are needed for the new system. The software is often identified first, as the hardware is often dependent on what software is needed. Only software that is needed to run the system should be specified. There may be different software identified for different types of user and process.

Once the software is known, the minimum hardware required to run that software can be identified. In addition to this, the amount of storage space required for the data being used by the system needs to be considered.

The system specification may plan for higher than minimum specifications, so that the system functions at a reasonable speed. These hardware specifications will include the processing power and the amount of memory required. External hardware components that are needed should also be specified and these should be based on the requirements of the user.

Design specification

The **design specification** is an illustration of how the system will look, what the data structures will be and how the system will function. It is intended to give the client an idea of what the system will look like, including the user interface, before it is developed, so that the client's feedback can be incorporated into the final designs.

You will look at the following parts of a design specification in more detail later in this chapter:

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- flowcharts
- data flow diagrams
- data collection forms
- screen layouts
- validation routines
- data dictionary.

In addition to this, a design specification could include:

- house style (logos, colours, fonts, styles, sizes)
- screen sizes
- connectivity diagram to show links between screens
- purpose of calculations.

Questions

- 1 Identify five stages in the system life cycle.
- 2 Explain why interviews are more often used than questionnaires for smaller groups of users.
- 3 State the purpose of the system specification.

Stage 2: Design

During the **design** stage, the overall structure of the system and details of the system components are designed. Diagrams can be used during the analysis stage to describe how a current system or process works, or diagrams can be used during the design stage to demonstrate how a new system will work. The required data storage is defined, including any databases or files used. Input forms and output forms are also designed during this stage.

System processing and flow of data through a system

Within a system, data is processed, for example organised or changed. Data is then passed, or flows, from one part of a system to another. It is important to document the flow and processing of data using diagrams.

Data flow diagrams

A **data flow diagram (DFD)** shows how data flows throughout a system. A DFD does not show the order of processes. The diagram uses different symbols to show the different elements as the data moves around the system (see Table 16.2).

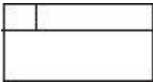
element	Purpose	Symbol
Data flow	This shows the data flowing from one part of the system to another.	
Process	This is an action that uses or manipulates data.	
Data store	This is a place where data is stored. For example, this could be a hard disk, cloud storage or a paper file.	
Data source or destination (inputs and outputs)	This is an external entity where the data originates or is intended to go. For example, this could be a customer or supplier.	
Duplication data source or destination	This is where the system has more than one source where data originates or is intended to go to avoid crossing data flow.	

Table 16.2: DFD elements and their symbols.

DFDs can exist at many levels: level 0 (context level), level 1 and level 2.

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At level 0, or context level, the diagram will show an overview of the whole system and the major data flows, processes and data stores.

WORKED EXAMPLE 16.03



Figure 16.6: The hotel booking system requires the guest to pay their hotel bill on arrival.

A hotel accepts bookings via email for its hotel rooms. Guests send a booking request using email, which is received by the hotel. The hotel then sends the guest a booking confirmation and reserves the room. When the customer arrives, they are presented with a bill for the room, which must be paid before they can begin their stay (see Figure 16.6). The DFD shows the system as the hotel booking system. It shows the guest as an external entity. It then shows the four items of data that flow between the guest and the hotel booking system.

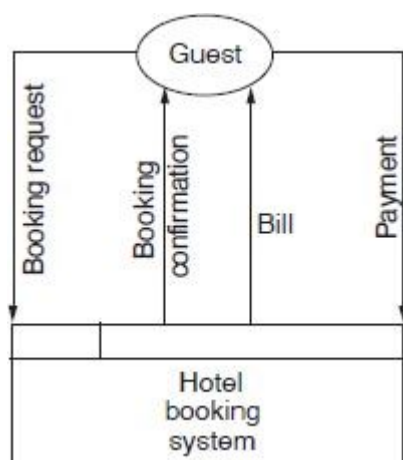


Figure 16.7: Level 0: DFD.

To create a level 0 DFD, identify the major external entities, along with the data flows between the external entities and the system. It is important to remember that it is the data flow that is being represented, not physical objects. Each data flow will be in one direction only.

UNPLUGGED ACTIVITY 16.02

Create a level 0 (context level) DFD for the following scenario.

A car rental company accepts bookings of cars by telephone (see Figure 16.8). A credit card payment for the deposit is taken from the customer at the time of booking. When the customer arrives to collect the car, they must provide details of

their driving licence, which are stored by the car rental company. The car rental company will provide the customer with details of the insurance and breakdown services for the car. The customer must pay the remainder of the rental cost before taking the car. The customer will be presented with an invoice showing the payments made.



Figure 16.8: A car rental company.

Level 1 data flow diagrams show the flow of data within part of a system by splitting major processes into sub-processes. For example, a hotel booking system might have a sub-process to check room availability. If the DFD is showing just part of a system, then other parts of the system are considered to be external entities. Each flow of data must be linked to a process, as the data has to be used before it can be stored or passed onto an external entity.

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WORKED EXAMPLE 16.04

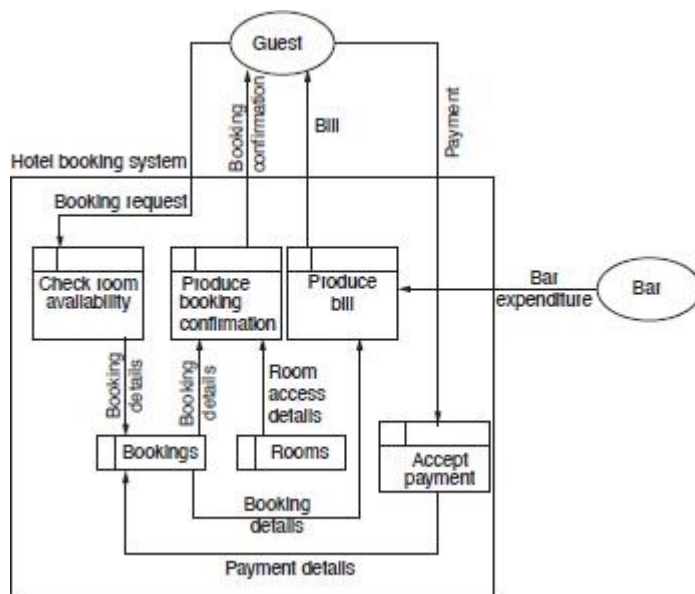


Figure 16.9: Level 1: DFD.

Figure 16.9 is a level 1 DFD for the hotel booking system. Any other aspects of the hotel system are considered to be external entities.

When the guest sends their online booking, the room availability is checked and the booking details are stored. When the guest arrives, their key card is activated by retrieving the room access details. When the guest is ready to leave, a bill is produced by retrieving the booking details and any bar expenditure from the bar system. When payment is made, the booking details are updated to confirm that payment has been made.

To create a level 1 DFD, first identify any external entities and any other parts of the system that will be classed as external entities. Identify the data flows to and from those external entities. Each data flow must have a process attached to it. A data flow cannot move directly from one external entity to another, from one data store to another, or between an external entity and a data store, because a process is required to use the data. DFDs focus on the data movement, rather than how the data will be processed.

UNPLUGGED ACTIVITY 16.03

Create a level 1 DFD for the car hire company introduced in unplugged Activity 16.02. Assume the DFD will be for the whole system.

Level 2 DFDs provide further detail about the various sub-processes within a level 1 DFD. Depending on the complexity of the system, level 2 DFDs may not be required. For example, a sub-process to produce a bill for a hotel guest might need data from a number of other systems, such as those within the hotel’s restaurant and pay-per-view TV.

System flowchart

A **system flowchart** shows the flow of data and processes within a complete system. A system flowchart helps explain how various elements of an information system are related.

DID YOU KNOW?

System flowcharts were popular in the 1960s and 1970s when data was stored on punched card (long, thin strips of card containing small holes) and magnetic tape (a reel of material that can be magnetised). Due to the complexity of modern systems, they are no longer commonly used.

Table 16.3 shows the symbols that represent the different elements within a flowchart.

Element	Purpose	Symbol
Input	Shows an input for the system, such as data entered using a keyboard.	
Output	Shows an output for the system, such as results displayed on a screen.	
Flow line	Shows the direction of flow.	

Process	An activity within the system, such as searching a database for certain data.	
Single document output	A single printed document.	
Multiple document output	Multiple printed documents.	
Magnetic disk file	Data stored in a file on a magnetic disk.	
Magnetic tape file	Data stored in a file on a magnetic tape.	
Display	Output on a visual display.	

Table 16.3: Flow chart elements and their symbols.

WORKED EXAMPLE 16.05

Figure 16.10 is a system flowchart for taking a hotel booking online.

Hotel bookings are input manually using a keyboard. While editing the booking, the data that has been input is displayed on the screen. Data for the guest who will be staying at the hotel is retrieved from the guests file. When the data has been input, it is validated and a process then saves the booking and prints a booking confirmation document that can be sent to the guest. Each morning, guest arrival sheets, which include information about each guest and the room they will be staying in, are printed. Also in the morning, a checkout list is printed that shows which rooms are due to be vacated that morning and therefore need to be cleaned.

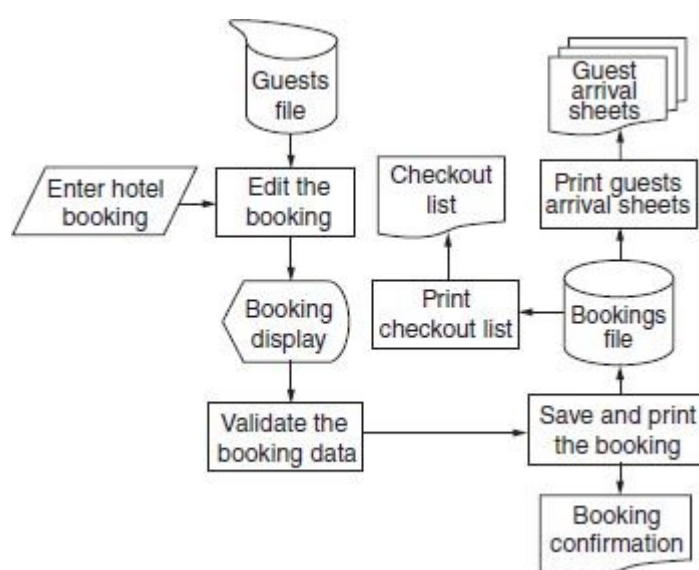


Figure 16.10: Online hotel booking flowchart.

UNPLUGGED ACTIVITY 16.04

Create a system flowchart for the following scenario.

A pizza delivery company accepts orders for pizzas by phone. The customer is asked for their address and the system checks the address is in the delivery file. The customer is asked for their order and the telephone operator inputs the order into the system. The order is added to the orders file. Payment details are then taken from the customer, which are saved to the payments file. A receipt is printed for the customer and an order sheet is printed for the chefs (see Figure 16.11).



Figure 16.11: Pizza boxes.

WORKED EXAMPLE 16.06

This is an example of a data dictionary for a table about students.

Attribute	Data type	Length	Format	Validation rule	Default value	Sample data	Primary key	Foreign key
Student ID	Integer	6	999999	Autonumber		459283	Yes	
Forename	Text	20	Xxxxx	Required		Sofia		
Surname	Text	20	Xxxxx	Required		Perez		
Tutor group	Text	3	99X			12A		TUTOR GROUP – TgId
Date of birth	Date	8	DD/MM/YYYY	<= 10 years before today		12/04/2012		
Date joined	Date	8	DD/MM/YYYY	>= 10 years after date of birth		01/09/2023		
Date left	Date	8	DD/MM/YYYY	> [Date Joined]				
Disability	Boolean	1		Required	False	True		

To create a system flowchart, identify the processes that take place within the system. Then identify the different files that will be used. Use flow lines to connect process to data files with the arrow pointing towards the data file if data is being stored, or the arrow pointing towards the process if data is being retrieved.

If user input is required, then add the input symbol at the appropriate place with the arrow pointing towards the process. Identify any documents that are produced by the system and link each one with a process with the arrow pointing from the process to the document.

Data storage

Databases

A data dictionary should be created to describe how data will be stored in tables within a database. The fieldnames should be identified along with their data type, field size and format. Primary keys and foreign keys should be identified, including the names of tables which the foreign keys link to. Any input masks, validation rules or default values should be identified for each field, along with an example of what typical data might look like. (You can find information about databases, including primary and foreign keys, in [Chapter 10](#).)

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UNPLUGGED ACTIVITY 16.05

Complete the following data dictionary for a table of vehicles. You may find it helpful first to review your understanding of data types, field sizes, primary and foreign keys and validation (see [Chapter 10](#)).

An entity relationship diagram should be created to show the relationships between the tables that will be used. (There is more detailed information about entity relationships in [Chapter 10](#).)

Attribute	Data type	Length	Format	Validation rule	Default value	Sample data	Primary key	Foreign key
Vehicle registration								
Make								
Model								
Engine size (cc)								
Transmission								
Number of doors								
Imported?								

Files (input and output)

Any files that will be used to import data should be designed, including the intended format of data within the file. Similarly, any files generated by the system should be designed, including the format in which the data will be exported. For example, data might be stored in a file as comma separated or tab delimited. The data stored within the file should also be specified, including the expected data type, length and format of each column.

Input forms

Features and elements of forms

Data collection forms are documents that are used to collect data without the use of a computer (see Figure 16.12). These could include membership application forms, questionnaires, job applications or reply slips. It is important to design the form in such a way that the required data can be collected.

Data collection forms typically contain several features and elements, such as:

- use a black font and any images should be in black and white, as the document may not be printed in colour and may not be visible to people completing the survey
- include instructions about how to complete the form
- give clear instructions about where the form should be returned
- identify which questions must be answered and which are optional
- provide enough space for each answer
- use tick boxes for multiple-choice lists
- make it clear how many options are allowed to be chosen from a multiple-choice list
- ensure all fonts are consistently used
- avoid cluttering the form with too much information or too many questions
- ensure the font style and size are legible
- if the respondent needs to complete a scale (e.g. 1–10), then explain what the scale represents (e.g. 1 = very dissatisfied, 5 = neither satisfied or dissatisfied, 10 = very satisfied).

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Membership Application

(PLEASE PRINT OR TYPE)

Company Name: _____

Primary Representative: _____ Title: _____

Address: _____

City: _____ State: _____ Zip Code: _____

Phone: () _____ Fax: () _____

Yes! Add me to your E-mail List: _____

Website Address (Free Link under category listing): <http://www.cheektowaga.org>

Yes! Website Advertising (Subject to Annual Renewal) \$25 _____
(Please include camera ready company logo or email HI RES file to: marketing@cheektowaga.org)

Number of Employees: Full Time _____ Part Time _____

Business Category (choose from list above): _____

Signature: _____ Date: _____

Referring Member & Company: _____

ANNUAL MEMBERSHIP INVESTMENT

Membership Classifications
If Your Business Employs...
(Part-Time employees count as 1/2 an employee)

☐ 1-2 Employees - \$155
(includes 1 representative)

☐ 3-20 Employees - \$195
(includes 1 representative),

☐ 21-49 Employees - \$295
(includes up to 2 representatives)

☐ 50-99 Employees - \$395
(includes up to 3 representatives)

☐ 100 or more Employees - \$495
(includes up to 4 representatives),

☐ Individuals - \$115 (will be listed by name only,
business name will not be listed),

☐ Community/Government Organization - \$115

☐ Banking Institutions - \$300

☐ Additional Category Listings
\$25 (per additional listing)

1) _____

2) _____

Please list any additional reps in this section if your
company employs more than 20 employees or your
Membership is at the \$295-495 dues level OR

☐ Yes — I would like to list additional representatives
at \$25 per rep/year

Name: _____ Title: _____

Email: _____

Name: _____ Title: _____

Email: _____

Name: _____ Title: _____

Email: _____

Membership rates subject to change.

METHOD OF PAYMENT

Dues Amount \$ _____

Application Fee \$ 40.00

Website Advertising \$ _____

Additional Reps.
or Categories \$ _____
(\$25 per each additional)

TOTAL \$ _____

() Cash

() Check Please make checks payable to:
Cheektowaga Chamber of Commerce

() MasterCard () Visa

Name as it appears on credit card _____

Credit card account number _____

Exp. Date _____

AGREEMENT: I hereby authorize payment to
be charged against the credit card listed above.

Authorized Cardholder Signature _____

Date _____

Cheektowaga Chamber of Commerce
2875 Union Road, Suite 50
Cheektowaga, NY 14227
Phone: 684-5838 Fax: 684-5571
www.cheektowaga.org

NOTICE
Membership dues in the Chamber of Commerce may be tax
deductible as an ordinary and necessary business
expense. Dues paid to the Chamber are not a charitable tax
deduction for federal income tax purposes. The
Chamber is not a charity, but serves as an advocate
organization for area business.

Figure 16.12: Example of a data collection form.

Appropriate use of forms for data collection

Paper-based forms are typically used when data collection can take place in a variety of locations, without the use of technology. Paper-based forms are usually quicker to create

than electronic forms, as detailed technical knowledge is not usually required to create the form. However, care needs to be taken to ensure the paper-based form can accurately record the required data to input into the system. It can also be time consuming to enter the data into the system, such as by a person manually typing the data on the form.

Electronic forms can enable data to be entered directly into the system, with the data viewed and analysed while being collected. Electronic forms can be programmed to ask the user specific answers based on previous responses, along with ensuring data entered is valid before being submitted. The use of electronic forms relies on the availability of technology and internet connectivity. This may be an issue for individuals that do not have access to modern technology (see [Chapter 6](#) The digital divide).

UNPLUGGED ACTIVITY 16.06

Explain how the application form in Figure 16.13 could be improved.

Membership Application Form

Name _____

Address _____

Gender _____

Age _____

Do you have any disabilities? Y/N

Which of the following types of movie do you watch? ☐ Horror ☐ Action
☐ Drama ☐ Comedy ☐ Other

Figure 16.13: Poorly designed data collection form.

Validation and checking of data collected by forms

Validation rules should be used wherever possible and be appropriate to reduce the number of possible input errors. Where possible, drop-down boxes should be used instead of lookup validation checks. For example, if a category needs to be selected, then a drop-down box should be used to select that category, rather than requiring the user to type in the category.

When designing a validation rule, identify:

- the input data that is to be validated
- the type of validation rule to be used
- the rule that will be used
- the error message that should appear if the data input is invalid.

Error messages should clearly guide the user as to what to do to correct the error.

Table 16.4 gives examples of validation rules for a range of input data.

Input data	Validation type	Validation rule	Error message
------------	-----------------	-----------------	---------------

Surname	Presence	Surname must be entered.	Please enter a surname.
Date of birth	Range	Date of birth must be at least 18 years earlier than today.	Applicant must be at least 18 years old.
Application number	Type	Must be a whole number.	The application number must contain only numbers.
Telephone	Length	Must be between 3 and 15 digits.	Telephone number must be between 3 and 15 digits.
Product code	Format	XX999XX9	The product code must be in the format XX999XX9 where X is a letter from A to Z and 9 is a number from 0 to 9.

Table 16.4: Examples of validation rules.

In addition to validation rules, any intended methods for verifying data input, such as visual checking, double data entry or hash control, should be specified. (You can find more detailed information about these verification methods in [Chapter 1](#).)

Input screen layouts

A screen can be used to ask the user to input data and to output information to the user, or a mixture of both. For example, Figure 16.14 shows information about a property that is for sale. It also allows the user to enter details about an existing property or add details about a new property.

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Figure 16.14: Example of an input screen.

When designing an input screen:

- Use colour carefully and appropriately. Different colours could be used for questions and responses or for different types of data. Colours that the user expects should be

used, for example the colour red to display a warning.

- Ensure all fonts are used consistently.
- Avoid cluttering the screen with too many elements, such as multiple text boxes, although it is helpful to fit all elements on a single screen if possible.
- Ensure the font style and size are legible.
- Include instructions about how to complete the form.
- Identify which questions must be answered and which are optional.
- Provide enough space for each answer.
- Use tick boxes for multiple-choice lists that can have more than one response.
- Use drop-down boxes (combo boxes) or option buttons (radio buttons) for multiple-choice lists that can only have one response.

(You can find more detailed information about the types of form controls that can be used on input forms in [Chapter 2](#).)

When designing an input screen layout, it is only necessary to indicate where questions and responses will go, the types of response options and the styles to use. The layout of any information should also be indicated. Figure 16.15 shows an example of a screen design. The way the input screen will be displayed is part of the next stage of the system life cycle.

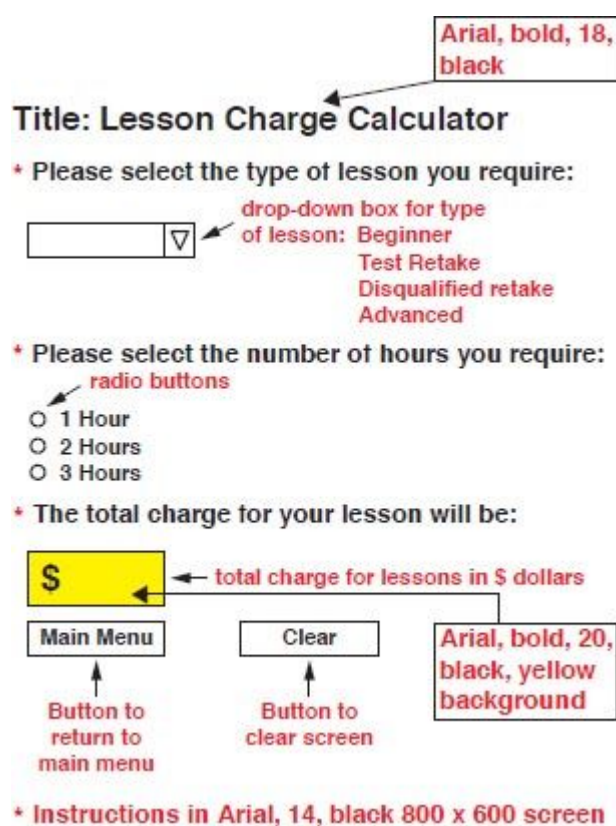


Figure 16.15: Example of a screen design.

Output reports

Output screen layouts

The design of output screen layouts is similar to input screens, except that there is no need for input elements, such as text boxes. However, controls may be required for the user to navigate information output, such as when viewing multiple results output from the system. For example, a scroll bar may be needed to move up and down large sections of text, or clickable buttons could be used to move between different screens.

Printed copy layouts

When designing a printed copy layout, as well as following the same principles for output screens, you need to consider the size of paper that will be used, the width of margins and the intended audience. Printed copies can often include tabular data.

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WORKED EXAMPLE 16.07

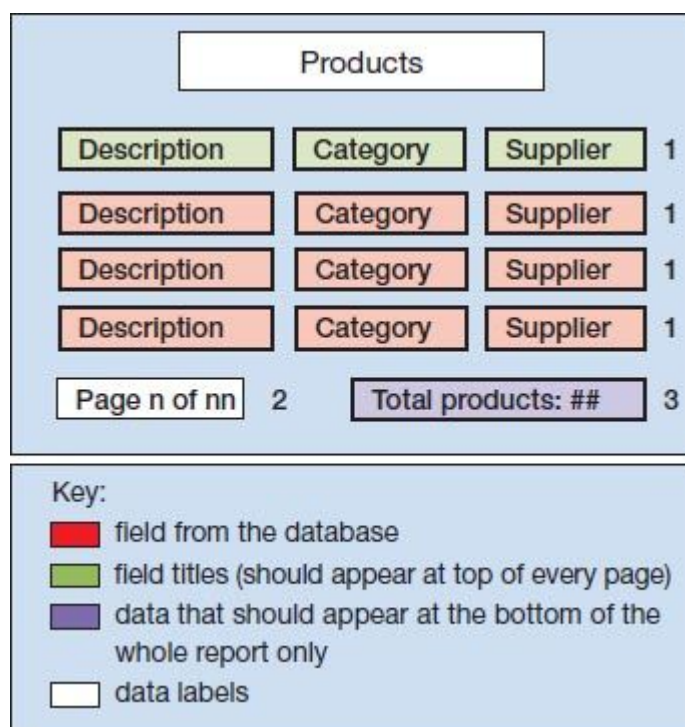


Figure 16.16: Example of report listing for products.

Figure 16.16 shows a report for listing all the products available.

- 1 The description, category and supplier data will be repeated for each record in the PRODUCTS table.
- 2 n = page number, nn = total number of pages.
- 3 ## = total number of products in PRODUCTS table.

In addition to these points, the size of paper, size of margins, font styles and sizes, colours and any gridlines should also be specified.

UNPLUGGED ACTIVITY 16.07

Design the layout of a report that will show a list of all the students in your class, including their dates of birth and contact details.

Questions

- 4 Identify the purpose of a data flow diagram (DFD).
- 5 Identify one rule for data flows within a level 1 DFD.
- 6 Describe the difference in usage between a tick box and an option button.

Stage 3: Development and testing

The **development** and testing stage involves producing each part of the system and ensuring it functions as expected. **Software developers** (see Figure 16.17) may use different methods of software development and these are discussed later in the chapter.



Figure 16.17: A software developer designs and creates computer programs.

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Test plans

Testing is necessary because software may contain **bugs** that cause the system not to function as expected. Bugs need to be found and fixed. It is important to ensure that the system is bug free, so that the users can use the system knowing that it will work reliably and behave as expected. Although it is challenging to ensure a system is free of bugs, a **test plan** can help to minimise the number of bugs by ensuring that all pathways through a system and examples of data input have been tested.

A test plan will identify the tests needed for each element of a system, such as when data is input or buttons are clicked. The test plan will include different types of test data, including valid, invalid and extreme, as outlined next and in further detail in [Chapter 8](#), so that inputs are thoroughly tested. Without a test plan, important parts of testing could be missed and bugs left undiscovered. A test plan will also cover the user's requirements and ensure that they are tested.

A good test plan will include all features and functionality to ensure the whole system is tested. The test plan will be continually updated to reflect any new areas to test as the software is developed.

As outlined in Worked Example 16.8, a test plan identifies what is being tested, the type of test, the input data that should be used to test it, the expected result of the test and space to record the actual result. Each test should be numbered.

WORKED EXAMPLE 16.08

The extract below, from a test plan tests the input of data where the date must be between 1/1/23 and 31/12/23 inclusive, and in the format dd/mm/yy.

The reason the test for 31/12/23 failed to be accepted may be because <31/12/23 was used in the validation rule rather than <=31/12/23. Similarly, the reason 31/12/23 failed to generate an error message may be because >= 31/12/23 was used in the validation rule rather than >31/12/23.

Number	Description	Type of test	Input data	Expected result	Pass/Fail
1a	Test the input of join date.	Valid	12/5/23	Accepted	Pass
1b		Extreme	1/1/23	Accepted	Pass
1c		Extreme	31/12/23	Accepted	Fail – error message displayed
1d		Invalid	15/7/22	Error message: the join date must be in 2023	Pass
1e		Invalid	4/6/19		Pass
1f		Extreme	31/12/23		Fail – accepted
1g		Extreme	1/1/23		Pass

Table 16.5: Extract from a test plan.

UNPLUGGED ACTIVITY 16.08

A parcel delivery company uses a number of codes to identify different types of parcel (see Figure 16.18). The code must start with an upper case letter between D and P (inclusive). Create a test plan to test the input of data for an upper case character to be input, which must be between the letters D and P (inclusive) in the English alphabet.



Figure 16.18: Parcel delivery.

In addition to testing input data, it is important to test that all calculations work as expected. Each input for a calculation will need to be identified and an expected result determined. Table 16.6 shows an example of how testing of calculations might be planned and executed.

Any links or buttons also need testing. Table 16.7 shows how a navigation button (main menu) and an action button (clear) would be tested.



Number	Description	Type of test	Input data	Expected result	Pass/Fail
2	Discount formula works for 2 hours	Calculation	Charge per hour = \$26 on quote worksheet	Test retake = \$50 in 2 hours column	Pass
3	Function for lesson charge	Calculation	Lesson type = advanced Number of hours = 2 on quote worksheet	Total charge = \$38	Fail = \$3.80

Table 16.6: Examples of test plan for calculations.

Number	Description	Type of test	Input data	Expected result	Pass/Fail
4	Main menu button	Button	Click on main menu button on quote worksheet.	The main menu worksheet opens.	Pass
5	Clear button	Button	Lesson type = advanced, number of hours = 2 Click on the clear button on quote worksheet.	Lesson type = (blank) Number of hours = (blank).	Fail – number of hours remained as 2

Table 16.7: Example of a test plan for buttons.

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Test data

As part of testing a system, a simulation of ‘live data’ has to be created for the purposes of testing. This is known as **test data**. Sufficient test data needs to be generated to ensure that the system will perform adequately when a large amount of data is entered. There will also need to be specific types of data to test different scenarios within the software, including validation rules and queries.

When testing the input of data that is to be validated, Table 16.8 shows the types of data that should be included as test data.

Type of test data	Description
Valid (also called normal data)	Data that will be accepted by the validation rule.
Invalid (also called abnormal or erroneous data)	Data that will not be accepted by the validation rule.
Extreme (also called extreme data)	Data that will be accepted by the validation rule, but it is at the limit of acceptability.

Table 16.8: Types of test data.

UNPLUGGED ACTIVITY 16.09

Select test data to test the input of numbers in the range 2500 to 5000.

Test data is also needed to test queries. Records will need to be created where there is data that meets the criteria of the query, does not meet the criteria of the query, only just meets the criteria of the query and only just fails to meet the criteria of the query. Where there is more than a single criterion, data should also be selected that only meets part of the criteria in case both parts have not been set up correctly.

WORKED EXAMPLE 16.09

The following data for records could be used to test the query for people in the job role 'engineer' and aged 50 or over.

Record number	Job role	Age	Reason
1	Engineer	65	Both criteria met.
2	Designer	25	Both criteria not met.
3	Engineer	25	Job role part met, age part not met.
4	Tester	65	Job role part not met, age part met.
5	Engineer	50	Job role part met, age part only just met.
6	Engineer	49	Job role part met, age part only just not met.

The more criteria that are used, the more records that will be required to test the query fully.

UNPLUGGED ACTIVITY 16.10

Data is stored about cars, including their make, model, registration number, transmission (automatic or manual), colour, distance travelled and year of registration. In a table, select test data that could be used to test the query to find all automatic transmission cars that were registered before 2016 (see Figure 16.19).



Figure 16.19: A database could be used to identify all cars of the same model in this car park.

The system will also be tested to see how it works under normal conditions and so a sample set of 'live data' will be used to test that the system functions correctly under normal conditions. 'Live data' is the actual data being used by the system once it has been implemented, and so a sample of 'live data' enables a simulation of a live system to take place. This type of testing with a sample of 'live data' will often be carried out by the end

users shortly before the system is implemented. The users will enter the data as if they were doing their normal job, simulating different processes that they would carry out.

Alpha testing and beta testing

Alpha testing is carried out by the developers, or a specialised team of testers, before a system is delivered to the user. This usually takes place close to the end of the development stage when the application is nearly ready for the user. Alpha testing can take a long time because each time a bug is found and fixed, testing has to be repeated to be sure other parts of the system are unaffected.

Beta testing takes place after alpha testing and is used when the software is almost complete. Beta testing takes place with a limited number of real users who have been selected to test the software. Alpha testing is planned and structured using test data to follow all pathways through the software, while beta testing involves customers using the software in a real-world environment, using real data. Where bugs are found within a beta version, new beta versions will be released for further testing before a final version is released.

White box and black box testing

Black box testing involves selecting input data and checking that the expected output data matches the actual output data, with no knowledge or understanding of what happens inside the black box. The black box could be the whole system or part of a system. **White box testing** involves the same process of input and output data, but the internal structure and logic of the program are known to the tester.

Table 16.9 describes the differences between white box and black box testing.

White box testing	Black box testing
Testing is carried out by the developers who made the software, as a detailed understanding of the system is required.	Testing can be carried out by specialist testers or, in the case of user testing, by the intended users.
Testing usually takes place with separate parts of the program. It ensures each module works in the way it was intended and, because the developers know how the system works, they can test pathways that a black box tester would not know about.	Testing usually involves testing the whole system. No knowledge of programming or the way the system works is required.
Testing will be focused on whether complex features, such as validation rules, have been developed correctly.	Testing will be focused on ensuring the requirements specification has been met.
Testing needs access system documentation.	Testing does not require knowledge of how the system was developed.
The tester can identify the type of test data that will be required, particularly related to valid, invalid and extreme test data.	The tester may not know what the boundary values are expected to be.
Each part of the program can be tested in depth.	Only a limited number of test scenarios are usually undertaken, as the tester is

White box testing	Black box testing
	unaware of how each part of the program works.
Test plans can be created for each calculation, navigation and input separately.	Test plans can be difficult to design, as many different input data options have to be considered, along with pathways through the system.

Table 16.9: Differences between white box and black box testing.

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Advantages and disadvantages of the different types of testing

Table 16.10 sets out the advantages and disadvantages of different types of testing.

Types of testing	Advantages	Disadvantages
Alpha	Provides early identification of critical issues. Testing takes place in a controlled environment.	May not reflect real-world usage. Repetition of testing can be time consuming.
Beta	Testing undertaken with a range of users. Real-world testing performed, include obtaining user feedback.	Limited control over testing process. Potential for loss of real-world data.
White box	Thorough coverage of all aspects of the system. Specific issues can quickly be identified.	Requires detailed technical knowledge of the system. May not identify issues experienced by typical users.
Black box	Identifies issues with the user experience. Ensures each part of the system integrates successfully.	Testing needs to be carefully planned to consider user inputs. Limited visibility of internal code may mean bugs are challenging to identify.

Table 16.10: Advantages and disadvantages of different types of testing.

Questions

- 7 Describe the purpose of using extreme test data.
- 8 Describe two differences between alpha and beta testing.
- 9 Explain black box testing.
- 10 Explain the importance of having a test plan.

Stage 4: Implementation

The **implementation** stage involves the system being installed within an organisation for the user.

Methods of implementing a system

There are four different methods of implementing (installing) a new system:

- parallel running
- direct changeover
- phased implementation
- pilot implementation.

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Parallel running

Parallel running is when a new system and an old system are run at the same time. On an agreed date, the new system will become live, but the old system will continue to run at the same time. Data will need to be duplicated from the old system to the new system, then any new data will be input into both systems, with output also produced from both systems. Parallel running will continue until the organisation is confident that the new system is working satisfactorily.

Direct changeover

Direct changeover is when a date is chosen for the old system to stop running and the new system to start running. The systems do not run at the same time and there is a clear break from the old system to the new system. Data will need to be transferred from the old system to the new system before the new system can be used.

Phased implementation

With phased implementation, parts of the new system will be introduced at different times. Phased implementation often takes place when there is a large system with functionality that can be easily separated into sections. The old system will run until a date that has been agreed, at which point part of the old system will be stopped and part of the new system will start. After a period of time, another part of the old system will be stopped and another part of the new system will start. Phased implementation will continue until all the new system is in place.

Pilot implementation

Pilot implementation takes place when part of an organisation starts to use the new system, while the rest of the organisation continues to use the old system. If the new system works successfully for the pilot implementation group, the system can then be implemented in other parts of the organisation. The pilot group may also be able to deliver training, or provide support, to the rest of the organisation when the system goes fully live.

Advantages and disadvantages of each implementation method

The most suitable changeover method will depend upon the circumstances surrounding a new system. Factors that will need to be considered include:

- how critical the system is to the running of the organisation

- cost of implementation
- number of users in the organisation
- the size and complexity of the new system.

The advantages and disadvantages of each method, and their suitability for a given situation, are shown in Table 16.11.

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Implementation method	Advantages	Disadvantages	Suitable when...
Parallel running	If the new system fails, this is the least risky because the organisation can continue to run using the old system. The accuracy of the new system can be tested against the old system and any bugs fixed.	Duplication of data input means additional costs. There may need to be additional hardware installed, as the old hardware is still being used, which will require physical space. Data may be input differently into the two systems, meaning that the data is not the same in both systems.	A system is critical to the operations of the organisation and any issues could have serious consequences. The performance of the old and new systems need to be compared. The old system and new system can be easily integrated.
Direct changeover	This is cost-effective to implement, as there is no duplication of systems or input data. The data being used will be consistent because it is only being used in one system at a time. There is no need for the new system to be compatible with the old system.	This is a risky method because a bug or bugs could lead to the system crashing (failing) with no backup. All the training will need to be done in advance of changeover and, if there are a lot of users, this could result in some forgetting what they have learned by the time they start to use the new system.	The old system is unreliable or does not meet the organisation's needs. A quick changeover is possible from the old system to the new system. Training and support are available to help users move to the new system immediately.
Phased implementation	Bugs can be identified and fixed within the part of the system that has	Delays can occur waiting for each phase to be running successfully before	The new system is large or complex, so it would be

Implementation method	Advantages	Disadvantages	Suitable when...
	changed, rather than the whole system. End users can be trained how to use each part of the new system and spend time using the new part, before being trained in the next phase.	the next phase can start. Users will be using both new and old systems, which could lead to confusion about how to complete a task and data being updated in the wrong system. Both the old and new systems need to be compatible for data to be used across both systems.	challenging to implement at once. The organisation wants to minimise the likelihood of any issues with the implementation. Users need time to learn about how different parts of the new system function.
Pilot implementation	Bugs found during implementation will only affect the pilot group that is using the system. Bugs found by the pilot group can be fixed before the system is installed for all users. The pilot group can train other users on how to use the system, as they will already be familiar with its operation.	This is a slower method of changeover because the rest of the organisation has to wait until the pilot has been completed satisfactorily. Users in the pilot group might be anxious about using the new system, as it may still have bugs, while users outside the pilot group may be disappointed that they have not been offered the opportunity to try the new software first. Both the old and new systems need to be compatible with each other for data to be shared between the pilot group and the users	The new system includes a number of new features, or is very different from the old system. The system needs further testing before being implemented fully. The new system requires a large amount of training.

Implementation method	Advantages	Disadvantages	Suitable when...
		still using the old system.	

Table 16.11: Advantages and disadvantages of implementation methods and their suitability in a given situation.

REFLECTION

How will you remember the different types of implementation methods and their suitability for different situations? Would you create a resource such as a mind map? Or is there another method you would use?

Questions

- 11 Identify four methods of implementation.
- 12 Describe one situation when direct changeover would be more appropriate than parallel running.
- 13 Describe one situation when pilot implementation would be more appropriate than phased implementation.

ACTIVITY 16.11

A number of computer systems have failed during implementation. Research some examples of major system failures, such as those relating to air traffic control systems (see Figure 16.20). Produce a list of what lessons could be learned from these failures.



Figure 16.20: Air traffic controllers use complex computer systems.

Documentation

As part of the implementation stage, documentation will be produced to provide further information about the system, for example a description of the hardware needed to run the system and how to operate certain features. Much of the content within the documentation

can be reused from documents previously produced, such as flowcharts and business requirements. However, some of the documentation, such as instructions for users and marketing teams, will involve writing new content.

Requirements documentation

The **requirements documentation** provides a reference point for both the system developers and users to ensure the system meets the intended purpose.

The requirements documentation outlines the functionality of the new system (how it will work) and includes:

- functional requirements describing the system's behaviour, including inputs and outputs
- non-functional requirements covering other aspects of the system, such as reliability, security and performance
- business requirements identifying the business goals and objectives the system needs to address
- system requirements outlining the hardware, software and infrastructure needs.

Design documentation

Design documentation provides an overview of the construction of the system. The design documentation includes an outline of the system architecture (the different parts of the system and how they fit together). It provides:

- an overview of the main system components, along with data flows
- designs for data storage, such as the structure of the system's database, and for the user interface, which defines the layout and design of the user interface and any forms used
- details of the software design, such as the algorithms used and the network design, which describes the network infrastructure elements.

Design documentation helps developers ensure the system's technical requirements are understood and guides the implementation of the system. The documentation may be referred to as a reference point by developers and users as the system is implemented and helps to support the maintenance of the system.

Technical documentation

Technical documentation is an overview of the structure of the system, how it was put together and how it works. It includes:

- a data dictionary to show how data has been structured within the system
- any programming code will be annotated to explain their purpose, along with a list of variables, including their datatypes and purpose
- all validation rules, with the criteria used for successful input and the error messages that they generate
- the purpose of calculations within the system and an explanation given of how each calculation works

- a list of all buttons and links, including where they are located and what their function is
- all files used by the system and their purpose
- flowcharts to show how different parts of the system work and other diagrams that may have been used during design and development, such as entity relationship diagrams and screen connectivity diagrams.

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The technical documentation will also include:

- an installation guide for teams installing the system, both initially and in the future
- all the results of testing, including any known bugs
- backup routines to show where files are stored, how the routines were configured and how to restore them from a backup
- security settings documented to show which groups have access to each part of the system and the permissions they have been granted
- list of software and hardware requirements.

Technical documentation is also needed so that anybody carrying out future maintenance on the system will be able to understand how the system was developed and how it is configured. It may be that the person carrying out future maintenance was not part of the original development team and so will not be familiar with the system without the technical documentation. Even if it is the same person or team carrying out the maintenance, they will need the documentation to remember the structure of the system.

User documentation

User documentation is a guide giving instructions to the user. It can be in electronic or printed format. A contents page should be included with page numbers and an electronic version would include hyperlinks to those pages. An introduction to the purpose of the user guide should be included. All the software and hardware requirements will be listed within the guide if applicable to the user.

The main part of the user guide will be the instructions on how to use the system. This should include written instructions, together with screenshots of the software and/or photographs of hardware. Screenshots and photographs can be annotated with labels. Bullets or numbering should be used to break down instructions into manageable tasks.

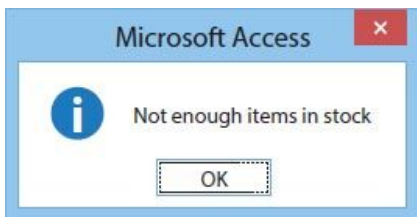
A glossary will show an alphabetical list of any technical terms that have been used within the user guide, along with a definition of each of those terms. There should be a troubleshooting section that includes a table of common problems and error messages, together with a description of what might have caused the problem and possible solutions for overcoming the problem. An index will be included at the end of the user guide with page numbers for each common term.

WORKED EXAMPLE 16.10

This is an example of a troubleshooting guide for a printer.

Problem	Cause	Solution
Orange light displayed on printer.	No paper in feeder tray.	Add paper to the feeder tray.
Red light displayed on printer.	Paper is jammed in the printer.	Open the paper feeder tray and check there is no paper stuck. Open the back door and check there is no paper stuck. Open the toner door, remove the toner and check there is no paper stuck. If any paper is found, gently pull any paper out and remove it.
Error message on computer says 'Printer Offline'.	Printer is turned off. Printer is not connected to the computer.	Turn on printer. Ensure the USB cable is connected between the computer and the printer.

Below is an example of a troubleshooting guide for an order processing system.

Problem	Cause	Solution
 Figure 16.21: Error message 1.	On the New order form, you have specified a quantity larger than the amount currently in stock.	Enter a smaller quantity.
 Figure 16.22: Error message 2.	On the Order details form you have ticked the Paid box before the invoice has been dispatched.	Check the invoice has been dispatched. If it has, then tick the Invoice dispatched box.

User documentation is needed so that the user can learn how to use the new system, or look up how certain features are supposed to work. The troubleshooting section will be important for the user to understand what has caused an error to occur and what they need to do to fix the problem and/or prevent the error from occurring again.

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UNPLUGGED ACTIVITY 16.12

Find a user manual for an electronic device or appliance at home. Identify the different sections that are used and compare them with those listed in

Worked Example 16.10.

Marketing documentation

Marketing documentation refers to the materials created to promote the system to potential users. Marketing documentation outlines the key features and functions of the system through providing an overview of the system’s capabilities. Marketing documentation may include product brochures or presentations, along with case studies outlining real-world examples of how the system has been used successfully. Content from other documentation or resources, such as screenshots and support materials, might also be provided to help understand the system.

Marketing documentation enables the benefits and features of the system to be described. Marketing documentation should highlight the main advantages of the system and the problems it solves. It may contribute to the success of the system by creating a positive image of the system for users.

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Advantages and disadvantages of the different types of documentation

Table 16.12 describes the advantages and disadvantages of the different types of documentation.

Documentation	Advantages	Disadvantages
Requirements	Provides an understanding of user needs and system functions. Reference point for both the developers and user.	Can be challenging to record all requirements accurately. Requirement changes can lead to delays and the document may need to be revised.
Design	Provides guidance for the development process. Gives an overview of technical aspects of the system.	May be time consuming to produce. Can be difficult to understand for end users.
technical	Outlines detailed technical information about the system. Helps with system maintenance and trouble shooting.	May be challenging for non-technical users to understand. May become outdated as the system develops.
User	Enables the user to understand how the system functions. Allows the user to trouble shoot common errors themselves.	User documentation may not address all tasks undertaken. Writing documentation for people unfamiliar with a system can be challenging.
Marketing	Highlights the key features and benefits of the system. Can create a positive image of the system.	Can be challenging to create engaging marketing materials. Materials may exaggerate the system’s capabilities, leading to

disappointed users.

Table 16.12: Advantages and disadvantages of different types of documentation

Questions

- 14 Give three sections you would expect to find in user documentation.
- 15 Describe the purpose of a glossary in user documentation.
- 16 Give a situation when technical documentation would be needed.

Evaluation

When a system has been developed and implemented, it will be evaluated. **Evaluation** is part of the implementation stage of the system life cycle. Evaluation will consider how well the system works, including its efficiency, how easy it is to use and whether it meets the intended use.

When evaluating a new system, the system's efficiency will often be examined through factors related to its use, such as speed of operation and reliability. Comparisons may be made with the old system to determine how well the new system is performing. An evaluation will also consider ease of use through examining feedback given and whether users can effectively undertake the required tasks. Finally, the evaluation will consider whether the system is appropriate for its intended use through examining the results of the evaluation. It may be necessary to make changes to the system, or implement new features, to ensure the system works as expected and can be used effectively.

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Evaluation techniques

Checking against specifications

During the analysis stage, specifications are produced for the user requirements, system and design. The content of these specifications is agreed by the developers and the organisation during the analysis, so checks can be made during the evaluation stage to ensure each part of the specification has been implemented. When parts of the specification have not been implemented as expected, further development of the new system might need to be undertaken to ensure the specifications are met.

Meeting user requirements

The evaluation will also check whether the system meets the user requirements. Each requirement will be considered in turn to determine whether it has been fulfilled. If a requirement does not meet user needs, there will be plans in place to rectify the situation. The efficiency of the new system will also be discussed, which could involve examining various performance measures, along with obtaining feedback from users, as outlined next. Should the new system not be as effective as the old system, further system development may be needed.

Feedback from users

As part of the evaluation, users can give feedback on the new system. Questionnaires may be given to users to find out their views of the new system and the impact on their work.

Selected users might be observed undertaking tasks to see whether the new system functions as expected and is being used in the intended manner. Some users will also be interviewed about their interaction with the new system. Users may be given an opportunity to make suggestions for future improvements or additions to the system.

Questions

17 State three elements that might be evaluated after a system has been installed.

Stage 5: Maintenance

Maintenance is the final stage of the life cycle. Maintenance takes place after a system has been completed and it is being used within an organisation. Four different types of maintenance are typically undertaken: perfective, adaptive, preventative and corrective maintenance.

Perfective maintenance

Perfective maintenance involves examining how to improve or enhance a system. While the system might be functioning as expected, there may be opportunities to improve a system's performance, or to undertake additional tasks. Sometimes, improvements might be possible due to new technologies becoming available.

If a system remains in place for several years without any improvements, then it may become outdated and inefficient. There may be newer, more advanced systems available. Users may also have new ideas about how to improve a system after having used it for a period of time.

For example, an online accounts application sends out automatic reminders to customers when payments have not been made. These reminders are sent to customers via email. However, a new technology has meant reminders can also be sent to a smart phone app, so perfective maintenance can be undertaken to integrate this technology within the existing system.

Adaptive maintenance

Adaptive maintenance focuses on modifying systems to ensure they remain functional and useable. Adaptive maintenance may be needed due to changes to internal procedures within an organisation. Adaptive changes may be required where the organisation has no control, such as the introduction of new government legislation. It is necessary to adapt to changes so that the system continues to work effectively and does not produce incorrect outputs. Adaptive maintenance is particularly important when alterations to a system are needed to ensure an organisation complies with the law. Adaptive maintenance may also take place as new technology is developed, such as a new operating system, web browser and hardware.

For example, a government introduced new requirements for organisations to provide pensions to all employees. The online accounts application needed to be updated to include a facility for checking that all employee payslips include pension payments, unless employees have opted out of the scheme. Adaptive maintenance was used to make the changes to the system and to ensure the company complied with the government's requirements.

Preventative maintenance

Preventative maintenance involves taking practical measures to stop problems arising within a system. Preventative maintenance is needed for both hardware and software to identify and address issues before they arise, which includes ensuring the system is kept secure. During preventative maintenance, systems might be updated, such as through using the latest drivers and ensure antivirus software is up to date. Hardware is cleaned and storage devices examined to check for errors and optimise disk space. Backup and recovery plans also form part of preventative maintenance, along with educating users on using the system appropriately, such as how to shut down the system correctly at the end of the working day.

For example, during preventative maintenance, an organisation's hardware is regularly cleaned to stop dust from blocking any fans, with regular scans of storage media carried out to ensure they are working properly. Data is also regularly checked for consistency and integrity, with the performance of the system monitored to ensure that the processor, memory and storage are all working efficiently.



Figure 16.23: Like cars, software needs to be regularly maintained to ensure good performance.

Corrective maintenance

Corrective maintenance ensures errors, bugs, or even system failures, are resolved so the system functions as intended. These errors need to be corrected quickly so there is limited impact on an organisation and users can continue to use the system. In particular, issues relating to system performance can be frustrating to users and have a negative impact on their use of the system (see Figure 16.23).

For example, a graphics application would intermittently stop responding for several seconds. Corrective maintenance was undertaken to identify the bug causing the application to stop responding, which was then fixed and users were able to use the software without interruption.

Advantages and disadvantages of types of maintenance

Table 16.13 lists the advantages and disadvantages of different types of maintenance.

Types of maintenance	Advantages	Disadvantages
Perfective	System functionality improves as new features are implemented and performance improves. Users' feedback can be used to make changes to the system.	Can require significant development effort, which can take a long period of time. Modifying existing code could introduce bugs or alter system performance.
Adaptive	Enables the system to remain compatible with technology and business requirements. Helps ensure the technology remains up to date and can be used in the future.	Integrating new technologies can be technically challenging. Changes might introduce compatibility issues, so thorough testing is needed.
Preventative	Helps prevent system failures and downtime. Avoids large costs associated with emergency repairs.	Ongoing costs are needed to carry out preventative tasks. Careful planning is needed to ensure tasks do not impact usage of the system.
Corrective	Addresses issues preventing the system working as intended. Issues are addressed immediately to minimise the impact on users.	Corrective maintenance needs to be reactive, which can be challenging and costly to implement. Users cannot use parts of the system while corrective maintenance is being completed.

Table 16.13: Advantages and disadvantages of different types of maintenance